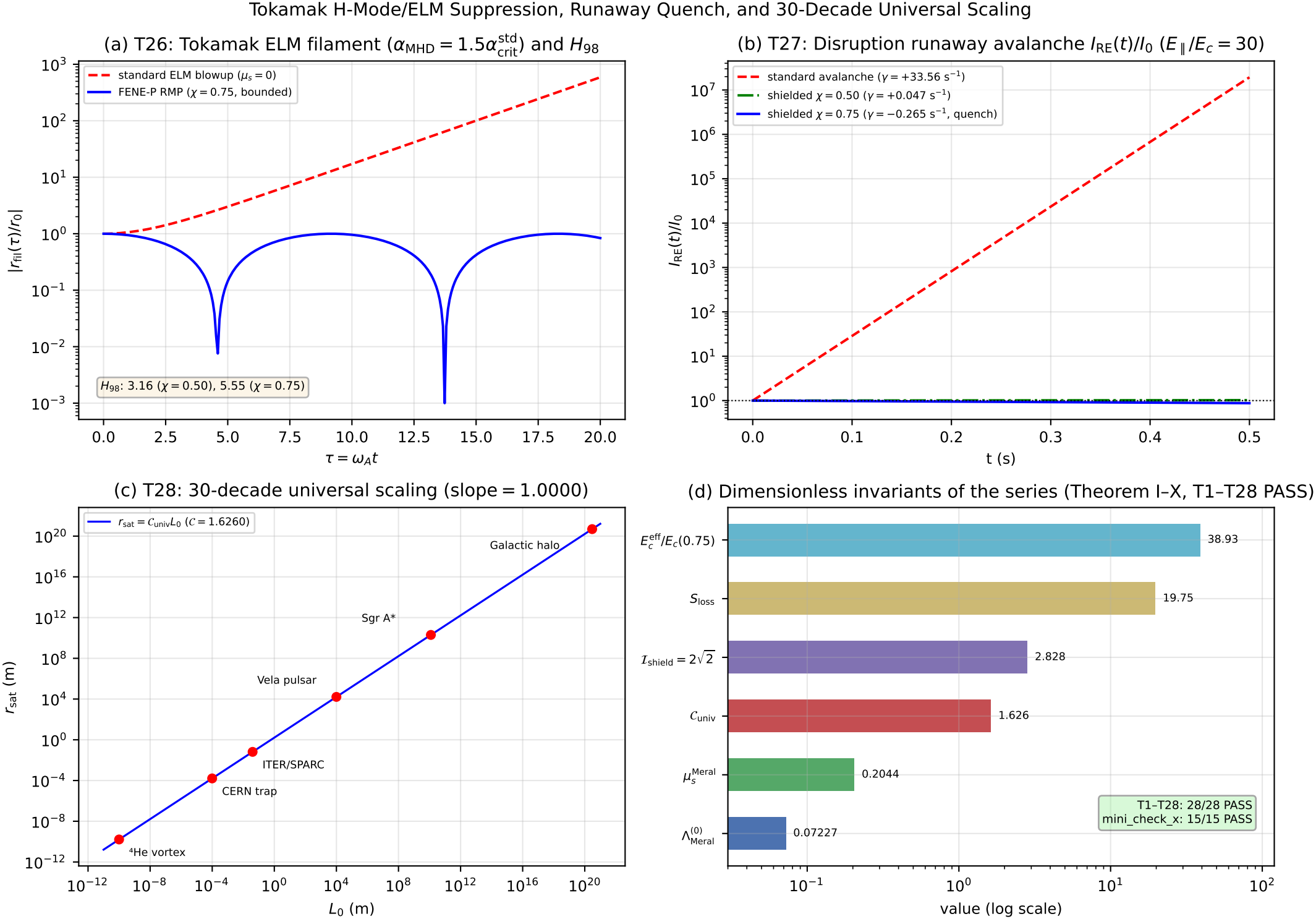
Tokamak H-Mode/ELM Suppression, Runaway Quench, and 30-Decade Universal Scaling
(a) T26: Tokamak ELM filament ($\alpha_{MHD} = 1.5\alpha_{crit}^{std}$) and H_{98}
standard ELM blowup ($\mu_s = 0$)
FENE-P RMP ($\chi = 0.75$, bounded)
H_{98}: 3.16 ($\chi = 0.50$), 5.55 ($\chi = 0.75$)
$|r_{fil}(\tau)/r_0|$
$\tau = \omega_A t$
(b) T27: Disruption runaway avalanche $I_{RE}(t)/I_0$ ($E_\parallel/E_c = 30$)
standard avalanche ($\gamma = +33.56$ s^{-1})
shielded $\chi = 0.50$ ($\gamma = +0.047$ s^{-1})
shielded $\chi = 0.75$ ($\gamma = -0.265$ s^{-1}, quench)
$I_{RE}(t)/I_0$
t (s)
(c) T28: 30-decade universal scaling (slope = 1.0000)
$r_{sat} = C_{univ} L_0$ ($C = 1.6260$)
Galactic halo
Sgr A*
Vela pulsar
ITER/SPARC
CERN trap
^{4}He vortex
r_{sat} (m)
L_0 (m)
(d) Dimensionless invariants of the series (Theorem I–X, T1–T28 PASS)
$E_c^{eff}/E_c(0.75)$
38.93
S_{loss}
19.75
$\mathcal{I}_{shield} = 2\sqrt{2}$
2.828
C_{univ}
1.626
μ_s^{Meral}
0.2044
$\Lambda_{Meral}^{(0)}$
0.07227
T1–T28: 28/28 PASS
mini_check_x: 15/15 PASS
value (log scale)